🧠 Test Plan: SMART-STOCK

Introduction

To validate that the application meets all documented business requirements, ensure strict role-based access, and confirm that the GUI is completely frictionless for non-technical users.

High-Level Test Plan and Scope

To ensure the SMART-STOCK system is robust and meets the exact needs of the hardware store, testing will be divided into the following phases:

- **Scope:** Testing will encompass the Admin and Staff access controls, real-time inventory deduction logic, and the automated demand forecasting algorithms.
- **Functional Testing:** Validating individual features against the Business Requirements Document (BRD).
- **Integration Testing:** Verifying that data flows correctly between different modules.
- **System Testing:** Evaluating the complete, integrated system for overall stability, performance, and security.
- **User Acceptance Testing (UAT):** Validating the solution with proxy end-users to guarantee the graphical user interface (GUI) is intuitive and distraction-free for non-technical hardware store staff.

Sample Test Cases

ID	Related Ticket	Test Case	Steps	Expected Result
TC-01	SS-13	Successful login with valid Admin credentials	1. Go to login page 2. Enter valid Admin email/password 3. Submit	User is authenticated and redirected to the Admin dashboard
TC-02	SS-13	Login rejected with invalid credentials	1. Enter incorrect password 2. Submit	Clear error message shown; user is not logged in

TC-03	SS-16	Prevent duplicate account registration	1. Register a new account using an email already in use 2. Submit	System blocks registration and shows a duplicate-email error
TC-04	SS-17	Inventory catalogue loads current stock data	1. Log in 2. Navigate to Inventory	Table displays SKU, name, category, price, and current stock count matching the database
TC-05	SS-18	Search returns correct filtered results	1. Type a partial product name in the search bar	List updates in real time to show only matching items
TC-06	SS-19	Add product rejects invalid input	1. Open "Add Product" 2. Enter a negative stock count 3. Submit	Form validation blocks submission with an error message
TC-07	SS-20	Low-stock alert triggers at threshold	1. Reduce a product's stock to its minimum threshold via a sale	Item is flagged/highlighted as low stock on the dashboard
TC-08	SS-33	Stock deducts automatically after checkout	1. Add items to cart 2. Finalize sale	Inventory quantity decreases by the sold amount immediately; transaction is logged with cashier ID

TC-09	SS-36	Staff account cannot access Admin-only pages	1. Log in as Staff 2. Attempt to navigate directly to an Admin-only URL (e.g. Supplier Records)	Access is denied / redirected; no Admin data is exposed
TC-10	SS-40	Stock-in updates inventory with correct unit conversion	1. Record a stock-in of 1 box (100 pcs) of screws 2. Check inventory count	Inventory reflects +100 individual pieces, tied to the supplier and timestamp

This is a living table — add a row per new Story before it enters a sprint, per the Development Readiness Checklist.

Bug Reporting

All bugs found during testing must use the standard Bug Report Template and be logged as a Bug-type issue in Jira, linked to the related feature ticket.